

NextGen RF Cognitive Radio

Final Presentation

5/6/2026

The Radioheads



AGENDA

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PROJECT OVERVIEW

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DELIVERABLES/TIMELINE

4

WORK COMPLETED

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FUTURE DEVELOPMENT



1

INTRODUCTION

Introduction

Team Members:

- Jack Miller – Team Lead
- Aaron Leeb – AI/ML Engineer
- Ethan Haugen – Embedded Systems & Hardware Engineer
- John Voracek – Data Pipeline & Visualization Engineer

Faculty:

- Dr. Flint Million – Faculty Coach
- Dr. Emily Gresbrink – Tech-Com Coach
- Other CS Faculty – SMEs/Resources

Our Client

Anne-Lyse Nono

Project Manager

Charles Viel

Lead Technical Specialist



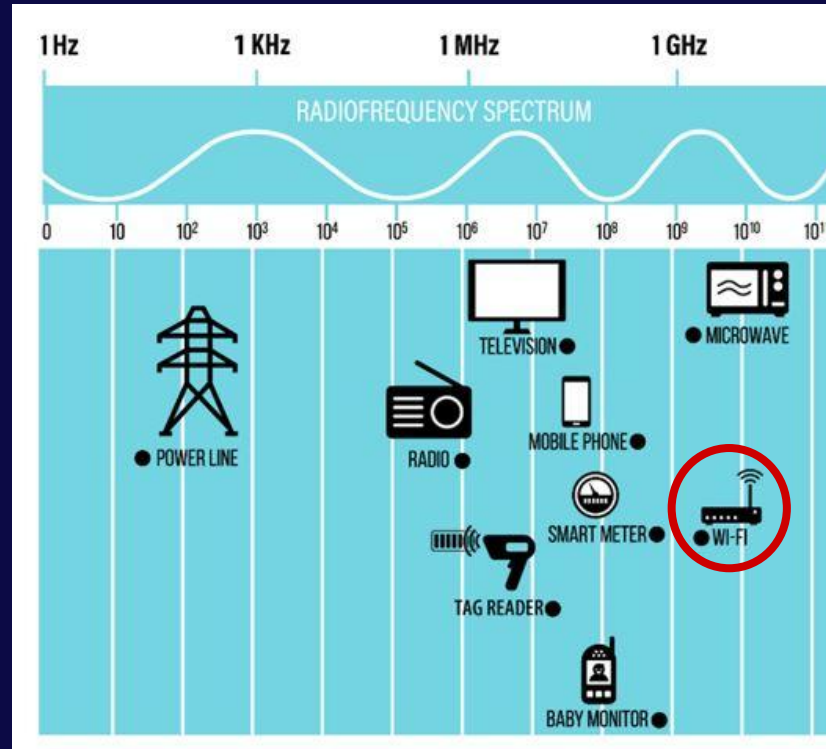
Designer and manufacturer
of various electronic
products, with a focus on
radio frequency devices.



2

PROJECT OVERVIEW

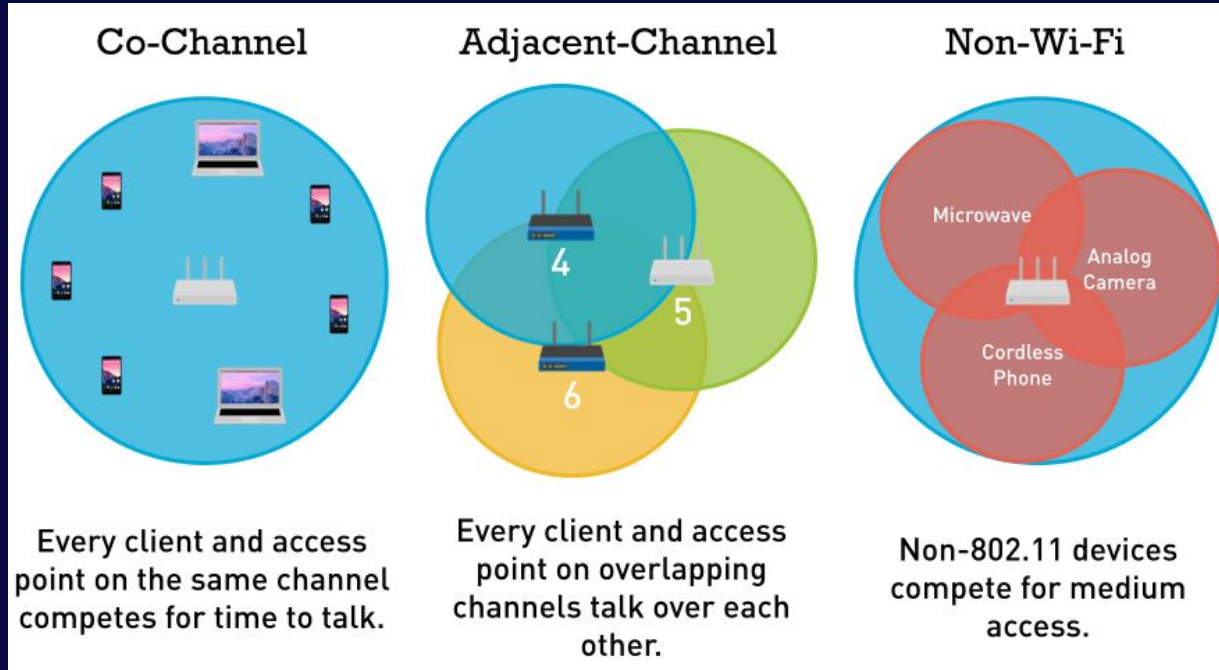
What are radio frequencies?



Our client faces issues with noise and interference on similar frequencies



There are three main types of interference



We designed a cognitive radio for interference mitigation

- Existing dynamic channel selection
- Monitor radio spectrum environment and make informed decisions to select “best” channel available



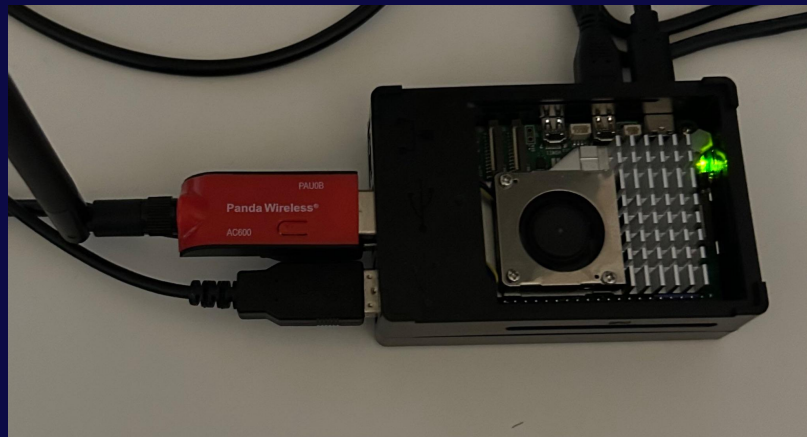
What makes our project different?

Modern AP



VS

Our Device



- Predicting channel quality before degradation
- Environmental / Temporal data
- Can scan the entire RF spectrum



3

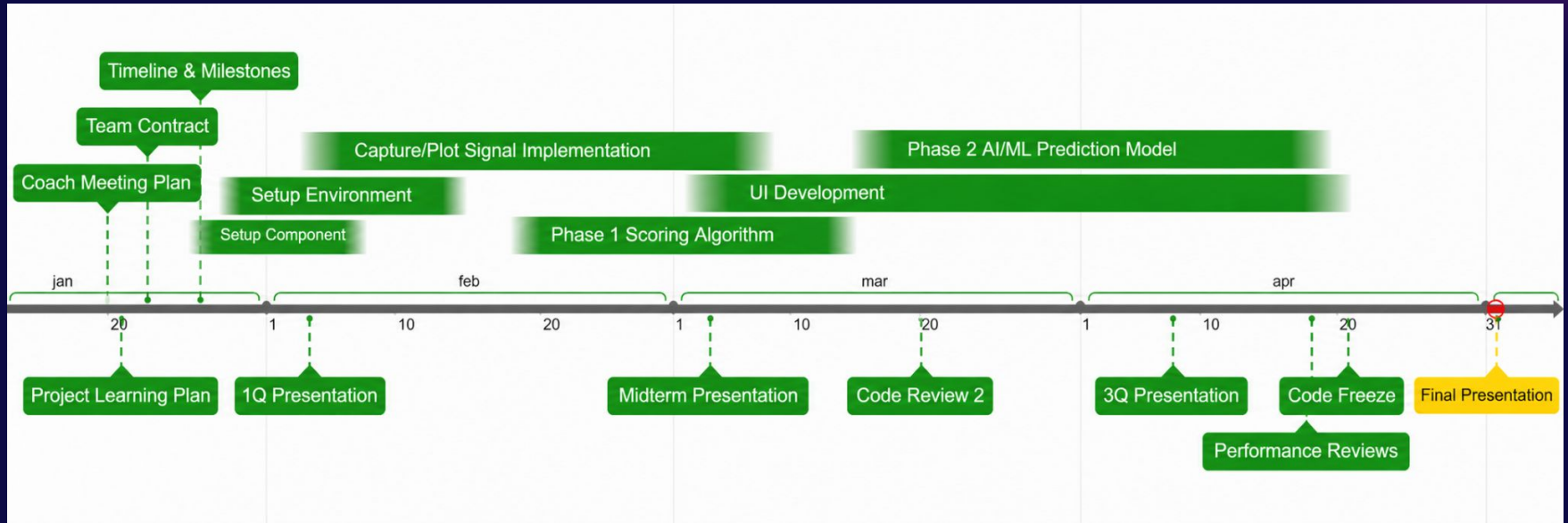
Deliverables/ Timeline

Deliverables

Status	Deliverable	Type of Work
	Setup Environment	Documentation, configure development environment
	Setup Components	Setup hardware and verify the compatibility of components
	Capture & Plot Signal Characteristics	Data extraction, processing, storage, and visualizing data
	Develop Prediction Algorithm	Research, develop scoring algorithm, train prediction model
	UI Development	Display data in comprehensive way that our client understands

	Completed		In Progress		Not Started
--	-----------	--	-------------	--	-------------

Timeline



Completed In Progress

Client Project



4

**Work
Completed**

Learning was a major part of our project

- Wi-Fi
- Raspberry Pi
- AI/ML

Hardware

- RaspberryPi
 - <https://www.raspberrypi.com/documentation/>
- Wi-fi Adapter
 - Wi-fi channel scanning - <https://www.netally.com/tech-tips/wifi-channel-scanning-a-guide-to-wifi-channels/>

Software

- OS
 - <https://ubuntu.com/download/raspberry-pi>
 - <https://documentation.ubuntu.com/desktop/en/latest/>
- Linux Wireless
 - <https://wireless.docs.kernel.org/en/latest/en/developers.html>
 - <https://www.geeksforgeeks.org/linux-unix/set-up-a-wireless-network-in-linux/>
- C Programming Language
 - <https://github.com/iczelia/C-Learning-Resources>
 - <https://www.learn-c.org/>
- Dev Environment
 - <https://learn.microsoft.com/en-us/windows/wsl/install>
 - <https://www.raspberrypi.com/documentation/computers/remote-access.html>

ML/AI

- <https://dyspan2025.ieee-dyspan.org/workshops>
- <https://scikit-learn.org/stable/>
- <https://www.turing.ac.uk/research/interest-groups/machine-learning-radio-frequency-applications>

UI Development

- <https://dash.plotly.com/>
- <https://realpython.com/python-gui-tkinter/>

Created an Implementation & Strategy Plan and discussed with our client

Deliverable 1 of 5

Set up environment

3. System Design

3.1. System Overview

We plan to implement a system composed of several key components working together in a pipeline:

1. A scanning module that gathers raw Wi-Fi data
2. A processing layer that extracts meaningful features from the data
3. A storage system to retain both raw and processed information
4. A scoring engine that evaluates channel quality in real time
5. A prediction component to predict future channel quality based on historical data and other factors
6. A visualization layer to display results and trends

The system will initially run entirely on the Raspberry Pi to simplify development and reduce dependencies on external infrastructure.

3.2. Data Collection Strategy

We will use the Linux `iw` tool to perform periodic scans of nearby Wi-Fi networks. These scans will provide information such as:

- Channel and frequency
- Signal strength (RSSI)
- Access point identifiers (BSSID/SSID)

Scans will be performed at regular intervals (approximately every 5–10 seconds) and parsed into a structured format for further processing.

5. Implementation Timeline & Anticipated Challenges

5.1. Timeline

The project will be developed incrementally over the semester:

- Early phase: environment setup, scanning, and data parsing
- Mid phase: data storage, feature extraction, and scoring algorithm
- Later phase: visualization and system refinement
- Final phase: machine learning integration and evaluation

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This phased approach allows us to build a functional system early and expand its capabilities over time.

5.2. Challenges

We expect several challenges throughout the project:

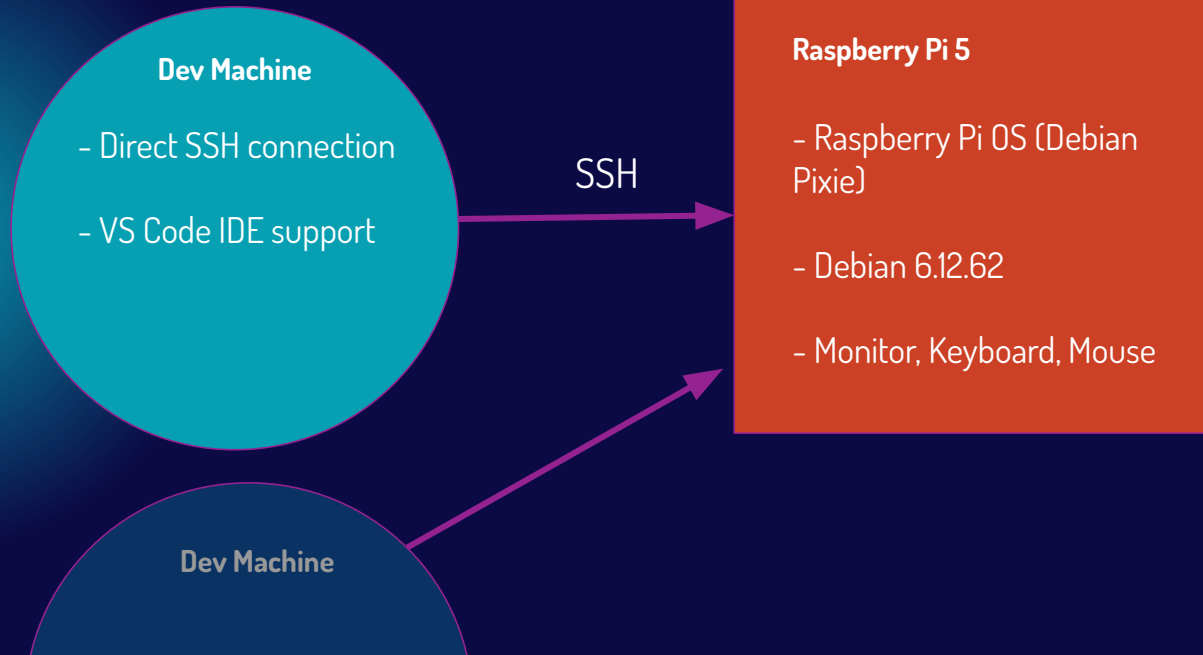
- Limited visibility into the full RF spectrum using standard scanning tools
- Hardware constraints of the Raspberry Pi
- Ensuring reliable and consistent data collection
- Balancing real-time performance with computational cost
- Lack of technical knowledge about RF and Wi-Fi science

We configured and documented our development environment

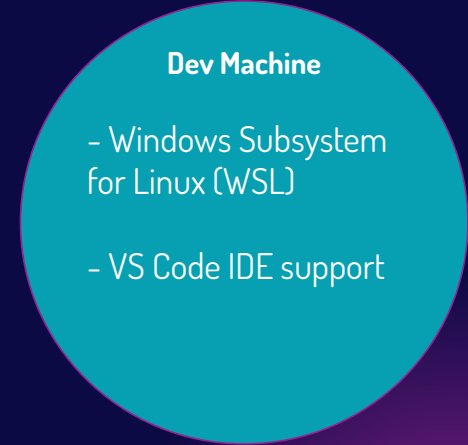
Deliverable 1 of 5

Set up environment

LAN



Remote



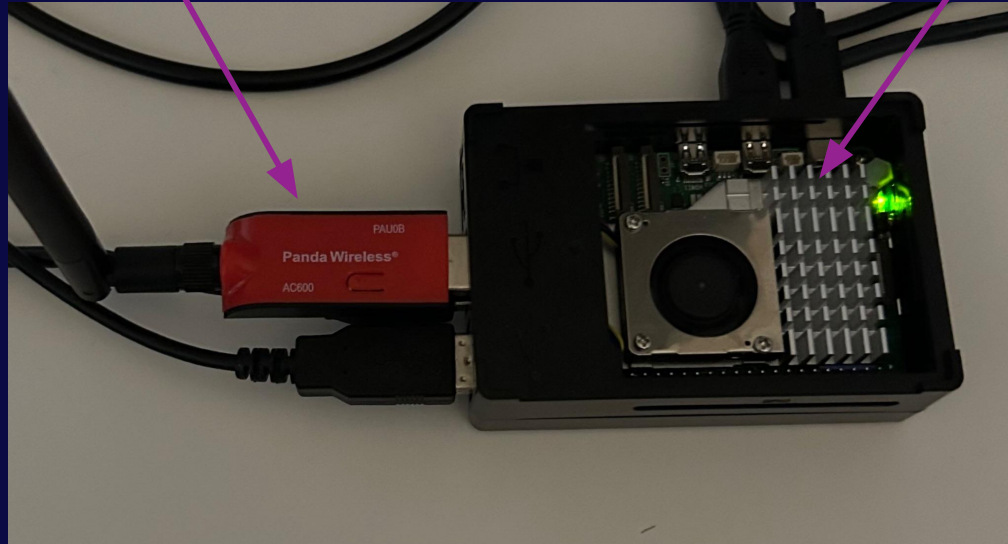
We successfully set up components

Deliverable 2 of 5

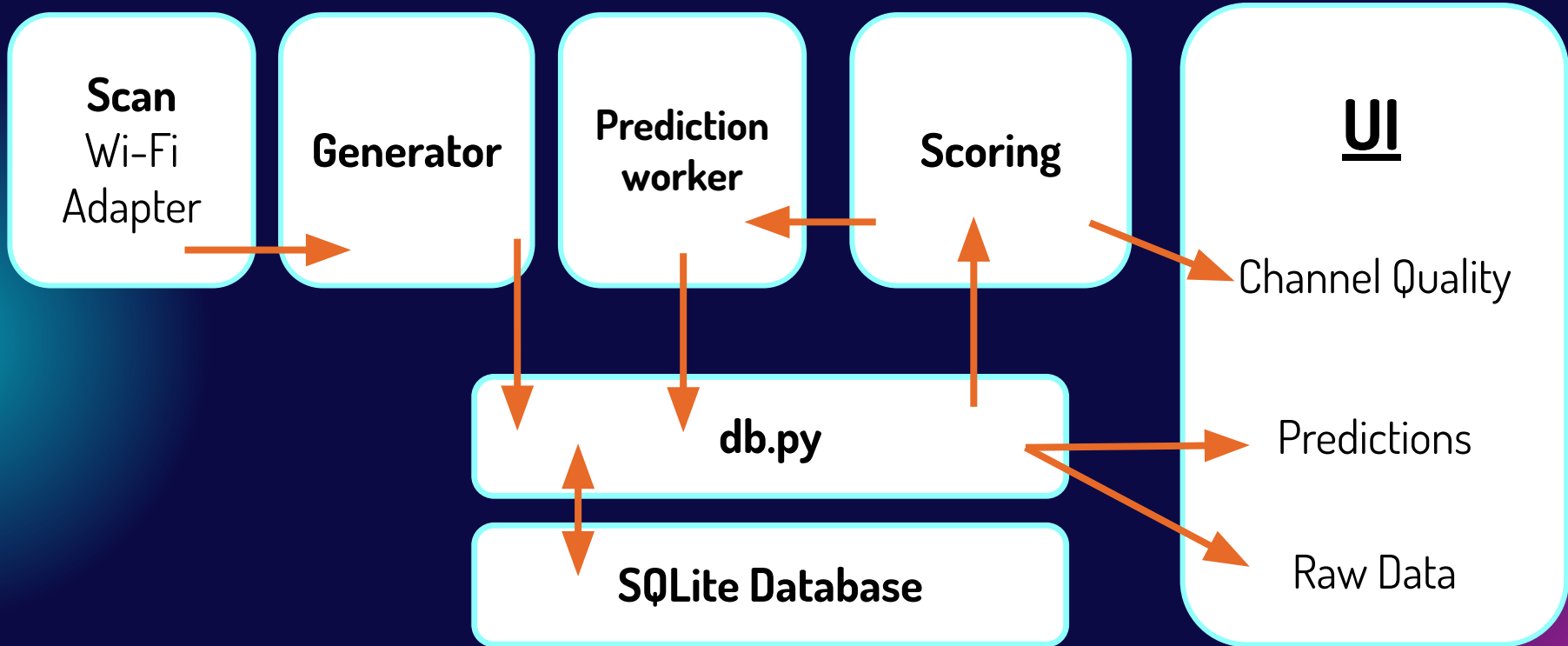
Set up components

Panda Wireless PAU0B (2.4–5GHz)

Raspberry Pi 5



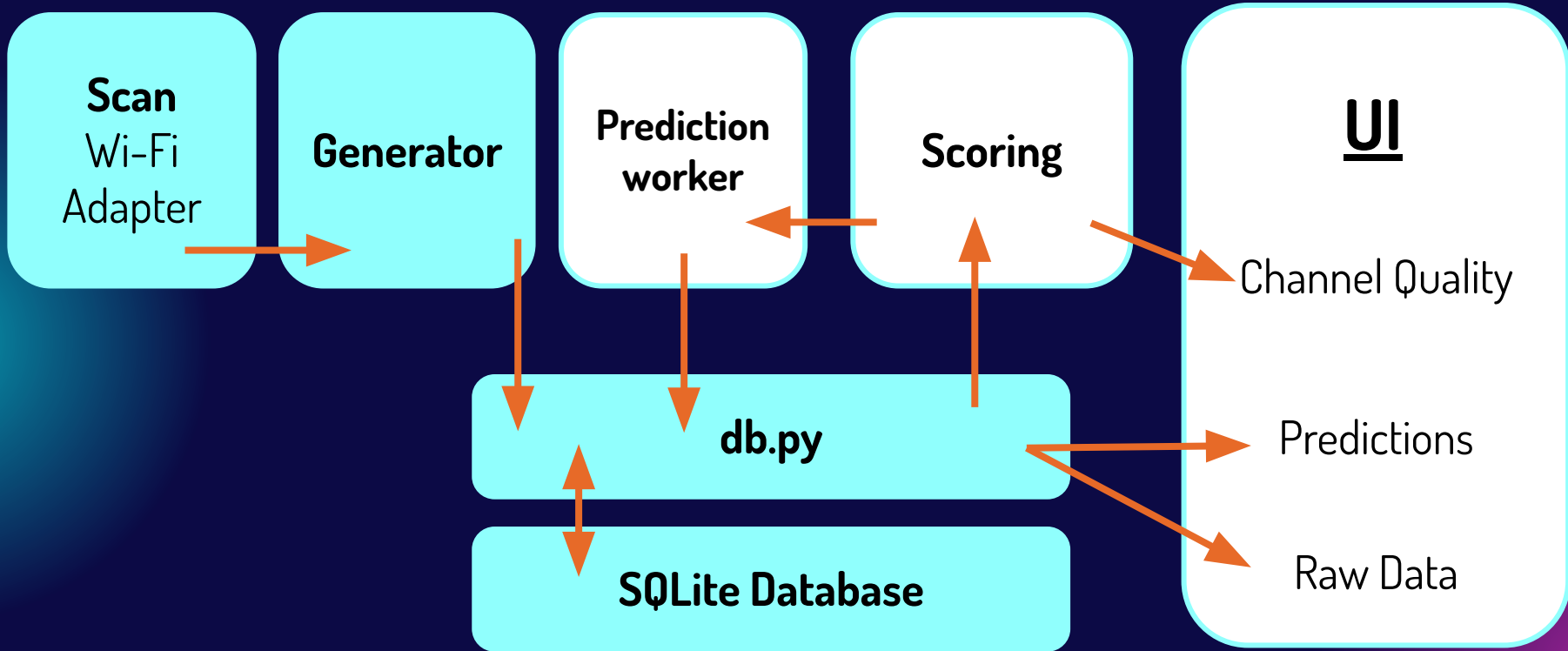
High level diagram of the final system architecture



First, we need to collect and store Wi-Fi channel data

Deliverable 3 of 5

Capture and plot
signal characteristics



We established what metrics we needed to collect

Deliverable 3 of 5

Capture and plot
signal characteristics

Channel Utilization

Percentage of active use of channel

AP Count

Number of visible access points

Strong Interferer Count

Number of APs with strong signal strength

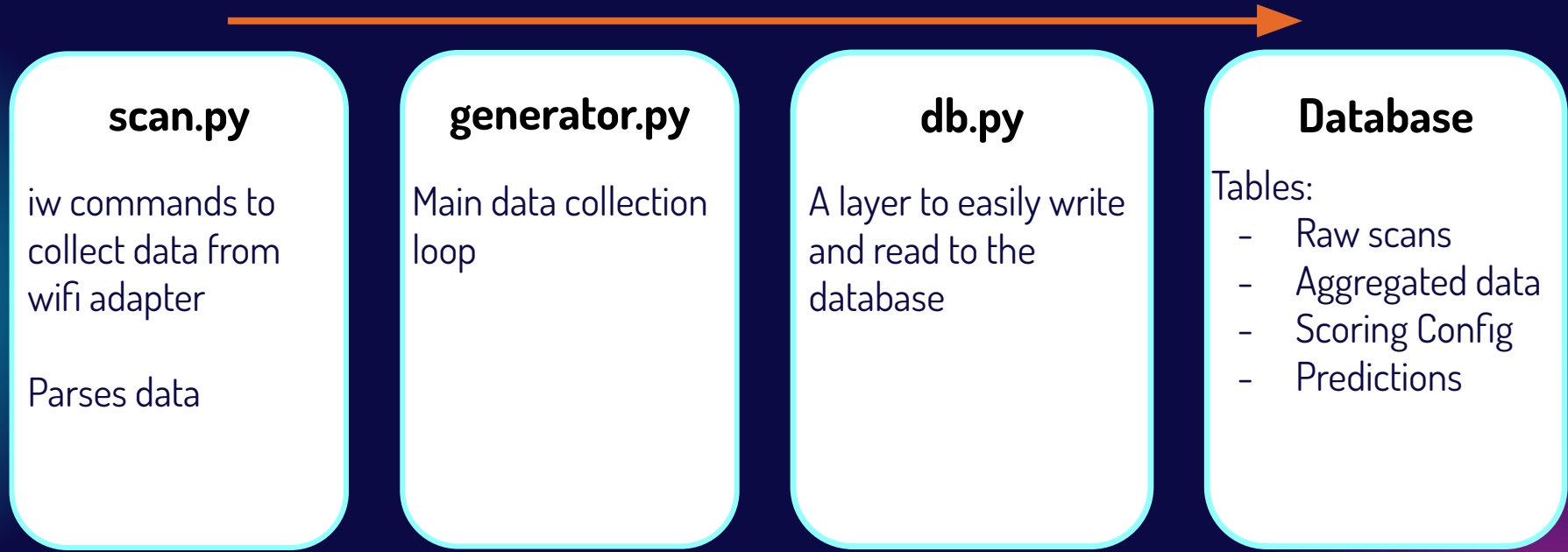
RSSI Distribution Summary

Average of the top three recorded signal strengths

Data collection pipeline

Deliverable 3 of 5

**Capture and plot
signal characteristics**



After data collection, we can assess channel quality and predict scores

Deliverable 4 of 5

Develop prediction algorithm

Scan
Wi-Fi
Adapter

Generator

Prediction

Scoring

UI

Channel Quality

db.py

Predictions

SQLite Database

Raw Data

The implemented scoring process visualized

Deliverable 4 of 5

Develop prediction algorithm



Scoring & prediction tools we are using

Deliverable 4 of 5

Develop prediction algorithm

Scikit-learn

- RandomForest Regressor model



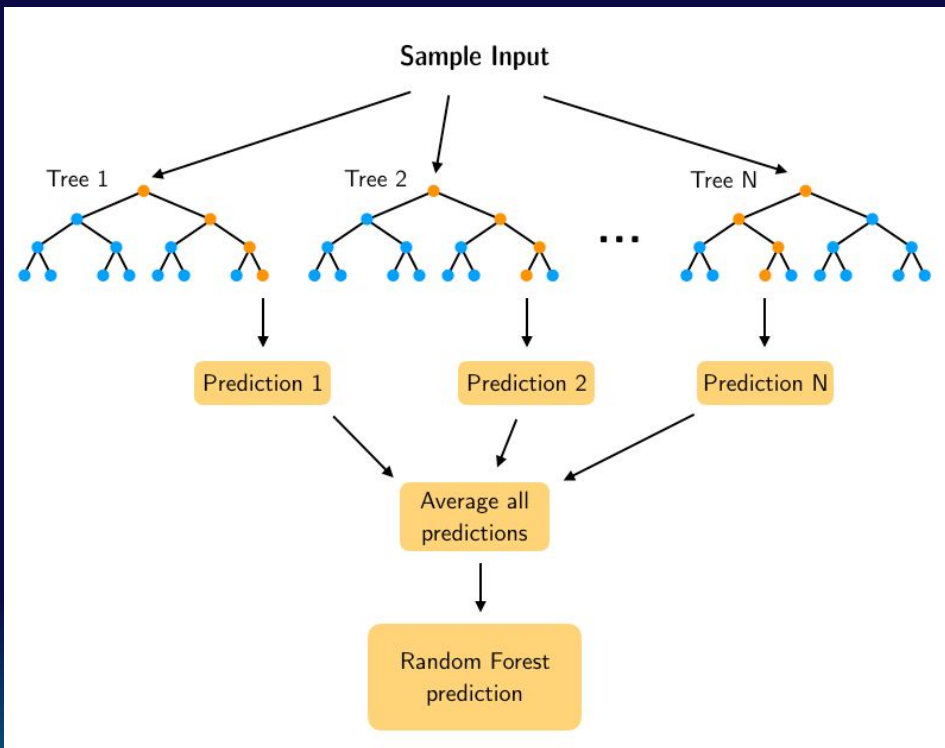
Pandas

- Data manipulation



We chose RandomForest Regressor for prediction

Develop prediction algorithm



Develop prediction algorithm

```
sample input:  
score_rollmean3 = 0.71  
score_rollmean12 = 0.67  
score_raw_mean = 0.74  
band_rank_lag1 = 4  
hour = 14  
...
```

```
is score_rollmean3 <= 0.677?  
no  
is score_raw_mean <= 0.823?  
yes  
is score_rollmean6 <= 0.741?  
yes  
=> predict 0.73
```

```
is score_rollmean12 <= 0.680?  
yes  
is band_rank_lag1 <= 10.5?  
yes  
is hour <= 15.5?  
yes  
=> predict 0.69
```

0.71

Prediction

60 trees

Develop prediction algorithm

We implemented mean average error and backtest.py to evaluate prediction accuracy

```

2026-05-03 15:16:53,220 INFO _main_: Prediction cycle complete: bucket=2026-04-03 15:25:00+00:00 channels=39
2026-05-03 15:16:53,225 INFO _main_: 5-min model MAE (holdout): 0.0086
2026-05-03 15:16:53,225 INFO _main_: Top channels for +5 min:
2026-05-03 15:16:53,228 INFO _main_: [2.4GHz]
2026-05-03 15:16:53,228 INFO _main_: #1 ch=1 predicted_score=0.6277 current_score=0.5974
2026-05-03 15:16:53,228 INFO _main_: #2 ch=11 predicted_score=0.5572 current_score=0.5380
2026-05-03 15:16:53,228 INFO _main_: #3 ch=2 predicted_score=0.5512 current_score=0.5402
2026-05-03 15:16:53,229 INFO _main_: [5GHz]
2026-05-03 15:16:53,229 INFO _main_: #1 ch=100 predicted_score=0.9144 current_score=0.9246
2026-05-03 15:16:53,229 INFO _main_: #2 ch=116 predicted_score=0.8415 current_score=0.8693
2026-05-03 15:16:53,229 INFO _main_: #3 ch=36 predicted_score=0.7914 current_score=0.8027
2026-05-03 15:16:53,230 INFO _main_: 10-min model MAE (holdout): 0.0104
2026-05-03 15:16:53,230 INFO _main_: Top channels for +10 min:
2026-05-03 15:16:53,231 INFO _main_: [2.4GHz]
2026-05-03 15:16:53,231 INFO _main_: #1 ch=1 predicted_score=0.6181 current_score=0.5974
2026-05-03 15:16:53,231 INFO _main_: #2 ch=11 predicted_score=0.5552 current_score=0.5380
2026-05-03 15:16:53,231 INFO _main_: #3 ch=2 predicted_score=0.5517 current_score=0.5402
2026-05-03 15:16:53,232 INFO _main_: [5GHz]
2026-05-03 15:16:53,232 INFO _main_: #1 ch=100 predicted_score=0.8958 current_score=0.9246
2026-05-03 15:16:53,232 INFO _main_: #2 ch=116 predicted_score=0.8218 current_score=0.8693
2026-05-03 15:16:53,232 INFO _main_: #3 ch=36 predicted_score=0.8143 current_score=0.8027
2026-05-03 15:16:53,233 INFO _main_: 15-min model MAE (holdout): 0.0115
2026-05-03 15:16:53,233 INFO _main_: Top channels for +15 min:
2026-05-03 15:16:53,234 INFO _main_: [2.4GHz]
2026-05-03 15:16:53,234 INFO _main_: #1 ch=1 predicted_score=0.6323 current_score=0.5974
2026-05-03 15:16:53,234 INFO _main_: #2 ch=11 predicted_score=0.5621 current_score=0.5380
2026-05-03 15:16:53,234 INFO _main_: #3 ch=2 predicted_score=0.5618 current_score=0.5402
2026-05-03 15:16:53,235 INFO _main_: [5GHz]
2026-05-03 15:16:53,235 INFO _main_: #1 ch=100 predicted_score=0.9062 current_score=0.9246
2026-05-03 15:16:53,236 INFO _main_: #2 ch=116 predicted_score=0.8347 current_score=0.8693
2026-05-03 15:16:53,237 INFO _main_: #3 ch=36 predicted_score=0.8126 current_score=0.8027
2026-05-03 15:16:56,993 INFO _main_: Stored prediction rows: 117

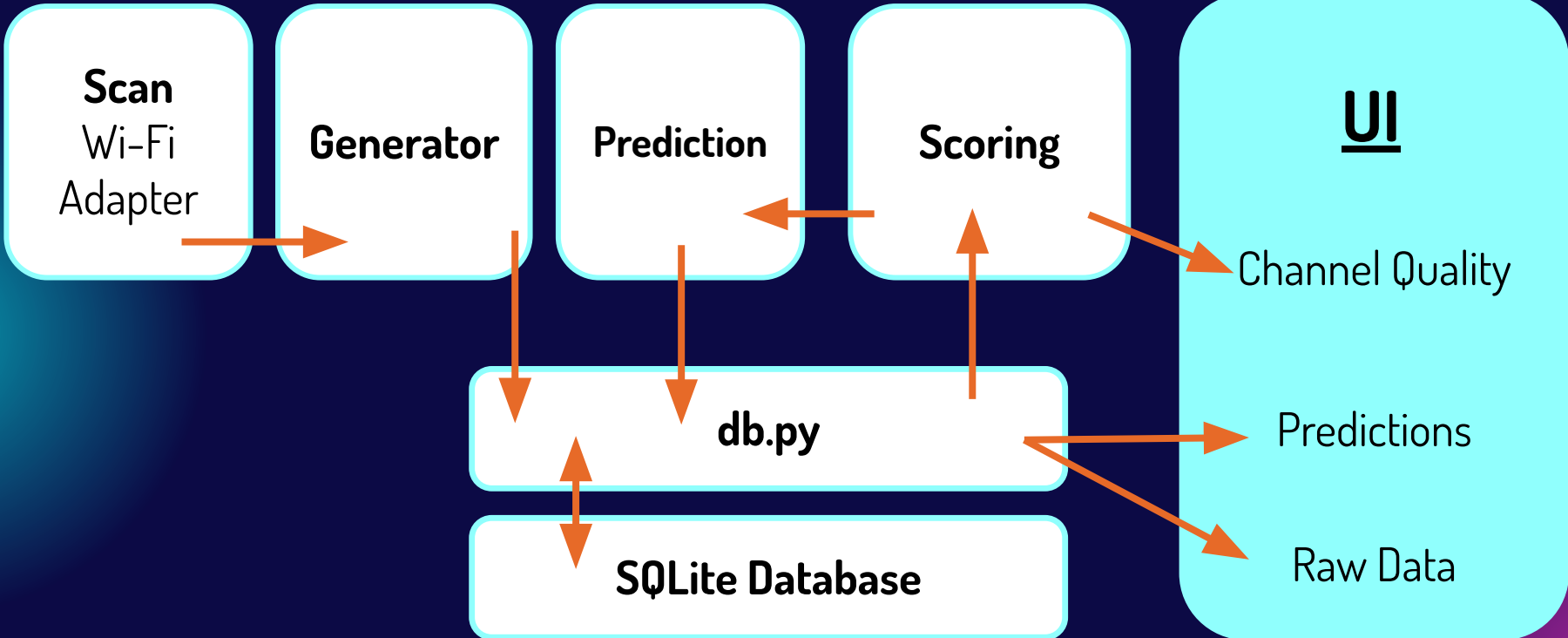
```

Terminal from predict_worker

Finally, we can visualize the data, scores, and predictions

Deliverable 5 of 5

UI development



The UI implementation is using Plotly Dash

Deliverable 5 of 5

UI development



We can view 5 GHz channels

UI development



We can compare channels

UI development



We can view raw historical metrics

Deliverable 5 of 5

UI development



We created an Admin page

Deliverable 5 of 5

UI development

Admin

WORKER PROCESSES

Scanner / GeneratorPID 3107

Prediction WorkerPID 5856

SCORING CONFIG

Changes are saved to the database and picked up by the scoring worker on its next run.

Chan Utilization		0.55	55%
RSSI Summary		0.3	30%
AP Count		0.1	10%
Strong Interferers		0.05	5%
EMA Alpha		0.4	0.40

Save Config

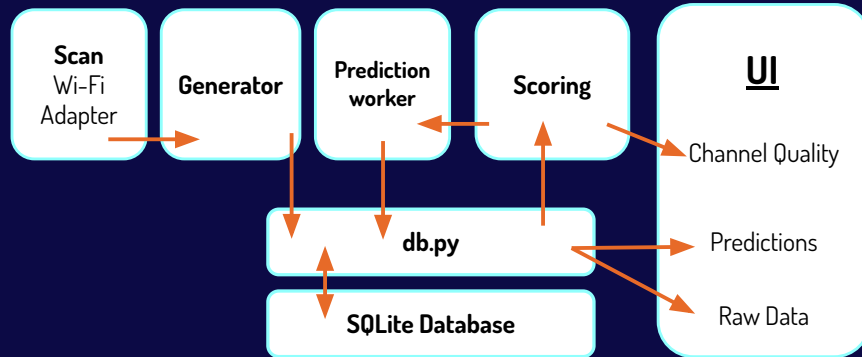
A decorative network graphic consisting of white dots connected by thin white lines, forming a complex web-like structure. It is located in the top right and bottom right corners of the slide.

5

SUCCESSES / CHALLENGES

Building a foundation for future projects was a success

Modular and decoupled architecture



Learning and Documentation

The image shows the cover page of a document titled "SYSTEM IMPLEMENTATION GUIDE" for the "NextGen Cognitive Radio Project". The document is from Minnesota State University, Mankato, Department of Computer Information Science, Computer Science Program. The title is "The Radioheads".

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• Hardware Setup Guide	5
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4. Project Setup Guide	8
• Repository Link	8
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• Cloning and Setting up the Project	8
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Page 1

The client is satisfied with our implementation

“I couldn’t have done better”

- Charles Viel

“I just wanted to express my gratitude and appreciation, you’ve been a really great team with great communication”

- Anne-Lyse Nono

Hi Jack and Team,

Fantastic work, the UI looks great!

Thanks,

-Anne-Lyse Nono
NextGen RF Design

Hardware incompatibility was a challenge while setting up components



Adapter | TL-WN722N

Tinjauan

Sp

<u>HARDWARE FEATURES</u>	
Interface	USB 2.0
Button	WPS Button
Dimensions (W x D x H)	3.7 x 1.0 x 0.4 in. (93.5 x 26 x 11mm)
Antenna Type	Detachable Omni Directional (RP-SMA)
Antenna Gain	4dBi
<u>WIRELESS FEATURES</u>	
Wireless Standards	IEEE 802.11n, IEEE 802.11g, IEEE 802.11b
Frequency	2.400-2.4835GHz

Understanding scoring metrics needed and Wi-Fi complexity was a challenge

SSID	BSSID	CHANNEL #	CHANNEL FREQUENCY		SIGNAL STRENGTH	SECURITY MODE
SSID	BSSID	CHAN	FREQ	RATE	SIGNAL	SECURITY
Jet City Nerds	fc:12:63:78:79:ed	1	2412 MHz		-42 dBm	WPA2/WPA3
(hidden)	9c:57:bc:c0:8e:a3	6	2437 MHz		-44 dBm	WPA2/WPA3
Grove200-131	9c:57:bc:c0:8e:a6	6	2437 MHz		-44 dBm	WPA2/WPA3
Jet City Nerds	fc:12:63:78:79:ee	157	5785 MHz		-49 dBm	WPA2/WPA3
Spectrum Mobile	de:12:63:78:79:ee	157	5785 MHz		-49 dBm	WPA2/WPA3
Grove200-234	9c:57:bc:e5:5d:86	6	2437 MHz		-56 dBm	WPA2/WPA3
Grove200-131	9c:57:bc:c0:8e:a7	108	5540 MHz		-56 dBm	WPA2/WPA3
(hidden)	d4:3f:32:e3:20:a3	6	2437 MHz		-56 dBm	WPA2/WPA3
Grove200-127	d8:8e:d4:08:ec:a6	6	2437 MHz		-58 dBm	WPA2/WPA3
Grove200-128	d4:3f:32:e3:20:a6	6	2437 MHz		-59 dBm	WPA2/WPA3

We received difficult but valuable feedback during our 1Q presentation

- Why do you think signal strength is important? Isn't the density (channel utilization) of traffic on that channel more important?
- Why do you think you're going to see time-series trends in channel behavior? Does the client have any evidence of this?
- How will you simulate heavy load on the channels? Are you going to be able to run in multiple environments so you don't overtrain for one environment?
- How will you compare the efficiency of your solution to previous solutions? How will you show you've done a good job?
- What happens if you have two cognitive radio algorithms running on the same radio network at the same time?
- Can you parallelize the work by starting on the AI/ML model early?

Feedback from 1Q

Only tested one model with the data, leaving prediction research incomplete

RandomForest was chosen over alternatives for several practical reasons suited to this deployment. It handles the mix of lag, rolling, normalized, and cyclical features without requiring feature scaling. It is robust to the sparse and occasionally missing data that characterizes early history windows. It produces bounded regression outputs naturally suited to the 0–1 score range. And it requires no hyperparameter search to produce reasonable baseline results.

Alternatives that could be explored by future developers include gradient boosting methods such as XGBoost or LightGBM, which typically outperform RandomForest on tabular time-series tasks but require more careful tuning and have higher inference overhead.

Didn't have the equipment to validate scoring algorithm and prediction in real-world scenario



Using wifi scanner to
control an AP and check
network speeds



```
Command Prompt

Reply from 8.8.8.8: bytes=32 time=10ms TTL=116
Reply from 8.8.8.8: bytes=32 time=9ms TTL=116
Reply from 8.8.8.8: bytes=32 time=9ms TTL=116
Reply from 8.8.8.8: bytes=32 time=9ms TTL=116
Reply from 8.8.8.8: bytes=32 time=11ms TTL=116
Reply from 8.8.8.8: bytes=32 time=13ms TTL=116
Reply from 8.8.8.8: bytes=32 time=9ms TTL=116
Reply from 8.8.8.8: bytes=32 time=10ms TTL=116
Reply from 8.8.8.8: bytes=32 time=9ms TTL=116
Reply from 8.8.8.8: bytes=32 time=9ms TTL=116

Ping statistics for 8.8.8.8:
    Packets: Sent = 33, Received = 33, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 8ms, Maximum = 13ms, Average = 9ms
Control-C
^C
C:\Users\ >
```



6

Future Development



Ideas for future development

- Implement dynamic channel switching
- Validate and fine-tune prediction algorithm
- Collect data in different environments



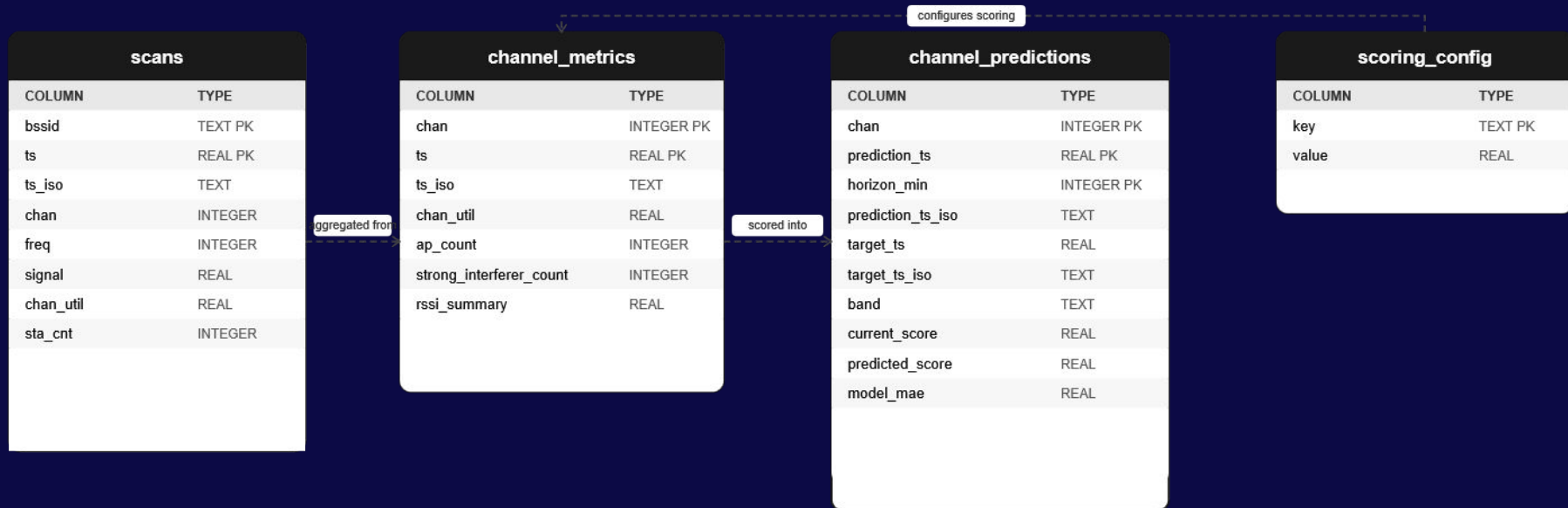


Thank you!
What questions do
you have?





Appendix



Our default weighted scoring algorithm for real time channel health

Deliverable 4 of 5

Develop AI/ML
algorithm

Score of channel = $0.55 * \text{Utilization} + 0.05 * \text{StrongInterferers} + 0.10 * \text{APCount} + 0.30 * \text{MaxRSSI}$

Communication was a success



Jack_Miller 4/9/26, 10:51AM

Hey @annelyse.n_93834, we made some progress this week on our prediction algorithm and UI. I was wondering if you'd be available to meet this Friday? Let me know if there is a time that works for you!



1



Client meetings:

**Wednesdays 11:00
– 11:30 AM**

MT

Miller, Jack T

To: Million, Flint D



Reply



R



Aaron Leeb - Performance Ev...

233 KB



Performance Evaluation CS S...

202 KB



VoracekJohn_S26_Performan...

238 KB



⌵ Show all 4 attachments (900 KB) ☁ Save all to OneDrive - Minnesota State ⬇ Download all

Hi Flint,

Attached are our team's performance reviews. Let me know if you have any questions or concerns!

Thanks!
Jack Miller

Coach meetings:

**Wednesdays
10:30 – 11:00 AM**

We captured Wi-Fi information for each detected access point using iw

SSID	BSSID	CHANNEL #	CHANNEL FREQUENCY	SIGNAL STRENGTH	SECURITY MODE	
radioheads@nextgen:~/wifi_scanner \$ python3 scan.py						
SSID	BSSID	CHAN	FREQ	RATE	SIGNAL	SECURITY
Jet City Nerds	fc:12:63:78:79:ed	1	2412 MHz		-42 dBm	WPA2/WPA3
(hidden)	9c:57:bc:c0:8e:a3	6	2437 MHz		-44 dBm	WPA2/WPA3
Grove200-131	9c:57:bc:c0:8e:a6	6	2437 MHz		-44 dBm	WPA2/WPA3
Jet City Nerds	fc:12:63:78:79:ee	157	5785 MHz		-49 dBm	WPA2/WPA3
Spectrum Mobile	de:12:63:78:79:ee	157	5785 MHz		-49 dBm	WPA2/WPA3
Grove200-234	9c:57:bc:e5:5d:86	6	2437 MHz		-56 dBm	WPA2/WPA3
Grove200-131	9c:57:bc:c0:8e:a7	108	5540 MHz		-56 dBm	WPA2/WPA3
(hidden)	d4:3f:32:e3:20:a3	6	2437 MHz		-56 dBm	WPA2/WPA3
Grove200-127	d8:8e:d4:08:ec:a6	6	2437 MHz		-58 dBm	WPA2/WPA3
Grove200-128	d4:3f:32:e3:20:a6	6	2437 MHz		-59 dBm	WPA2/WPA3

Documentation has been a success

My files > NextGen RF Spring 2026

	Name
	Meeting Notes
	Coach Meeting Plan.docx
	MSU CS Project Scope S26- Predictive Mo...
	Project Learning Plan.docx
	Project Roadmap.docx
	Team Contract.docx
	Timeline and Milestones.docx

Cognitive Radio Channel Selection — Technical Findings & Design Rationale

1. Project Goal

The system aims to determine the optimal Wi-Fi channel for operation by analyzing real-time RF conditions and historical trends. The solution combines wireless scanning data, channel activity measurements, and machine learning-based prediction to rank channels according to expected performance.

The design must operate on a Raspberry Pi using a USB Wi-Fi adapter and rely on passive environmental observation without disrupting network operation.

2. Data Collection Strategy

Two primary Linux wireless tools were evaluated:

2.1 Passive Scan Data (iw scan)

Provides information about detectable access points:

- Channel number
- RSSI (signal strength)
- BSSID and SSID
- Optional metrics (utilization, station count when supported)

Strengths:

- Identifies co-channel network presence

Documentation has been a success



MINNESOTA STATE UNIVERSITY, MANKATO
DEPARTMENT OF COMPUTER INFORMATION SCIENCE
COMPUTER SCIENCE PROGRAM

HANDOVER REPORT NextGen Cognitive Radio Project "The Radioheads"

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Handover Document

README

wifi_scanner

WiFi Scanner is a Linux/Raspberry Pi data collection tool that captures nearby Wi-Fi conditions over time and stores them in SQLite for later analysis and channel recommendation work.

It continuously combines:

- AP-level scan data from `iw dev <iface> scan -u`
- Channel-level utilization counters from `iw dev <iface> survey dump`

The project also includes tests and a scoring notebook for backtesting channel selection logic. It now also includes a lightweight prediction worker that refreshes every 5 minutes using scored channel history from `channel_metrics`.

What This Project Does

On each scan cycle, the collector stores:

- Raw AP observations (BSSID, channel, frequency, signal, BSS load utilization, station count)
- Per-channel metrics (channel utilization, AP count, strong interferer count, RSSI summary)

Channel scores are computed just-in-time from `channel_metrics` on each request (UI, prediction worker) and are not persisted separately.

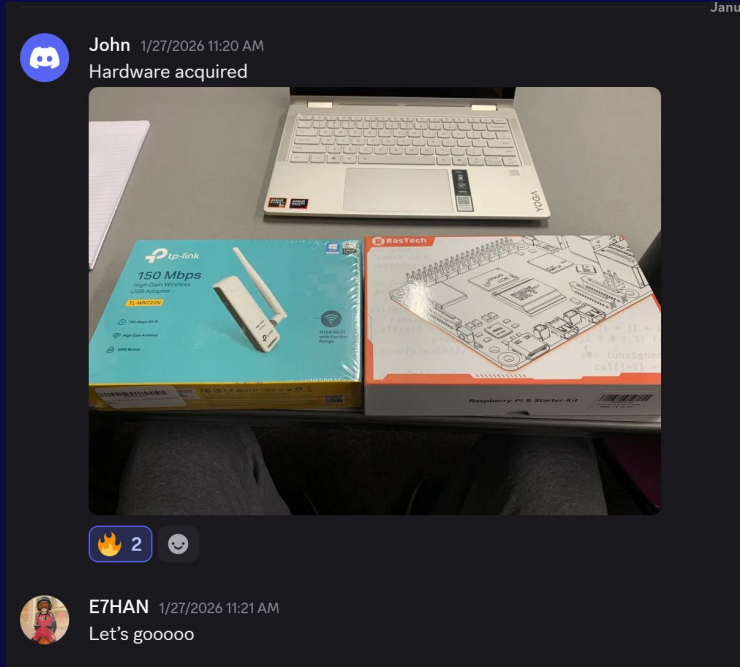
Repository README

docs

- modules.md
- overview.md
- predict_worker.md
- scans.db
- scoring_notebook.md
- scoring.md
- testing.md

Documents in Repo

Our team chemistry has been a success



We reevaluated our key channel quality metrics

Channel #	Timestamp	Channel Utilization	Access points on channel	Access points with high RSSI	Signal strength
chan	ts_iso	chan_util	ap_count	strong_interferer_count	rss_summary
1	2026-02-27T18:52:06Z	20.427435387674	1	0	-81.0
6	2026-02-27T18:52:06Z	84.1269841269841	6	5	-52.0
11	2026-02-27T18:52:06Z	31.8181818181818	4	4	-65.0

Scoring the stored channel metrics

Deliverable 4 of 5

**Develop AI/ML
algorithm**

Normalization

Processes the raw stored channel metrics

Converts metrics to a value from 0 - 1

Weighted Average

Normalized values get average out based on weights set in the scoring configuration

Final output is smooth with EMA

scoring.py

We defined an early data pipeline

Deliverable 3 of 5

Capture and plot
signal characteristics

Scan

Extract & Store

Clean

Score & Predict

Display

Our initial data collection was based on client-provided resources

Deliverable 3 of 5

Capture and plot
signal characteristics

3.2 Scanner mode

To perform analysis of the surrounding WIFI channels.

```
$ sudo iw wlan0 scan
```

```
BSS 88:9e:68:1c:84:88 (on wlan0)
....
freq: 2412.0
signal: -49.00 dBm
....
SSID: gilbel 2,4G
DS Parameter set: channel 1
....
BSS e6:bf:fa:87:8d:cc (on wlan0)
....
freq: 2412.0
signal: -64.00 dBm
....
SSID: zioutik
DS Parameter set: channel 1
....
```

WIFI FAQ

